Reproducible Four-Round Midori-64

Differential Characteristic Analysis

Egemen Çeken
Technical Research Report

October 2026

Abstract

This technical report presents a reproducible mixed-integer linear programming (MILP) analysis of four-round Midori-64. The S-box activity and exact differential transitions, the ShuffleCell permutation, and the MixColumn layer are encoded using a single consistent modeling framework adapted from Ilter and Selçuk. The complete four-round model has 1,280 binary variables and 2,049 constraints. An exhaustive component check verifies that the reduced 19-inequality S-box formulation accepts exactly the 97 valid input/output/probability patterns among all 1,024 binary patterns. Gurobi 13.0.3 *certifies* that the minimum number of active S-boxes is 16, with an objective value and best bound both equal to 16 and a zero optimality gap. An independent word-level dynamic program reproduces the same four-round bound without MILP. A feasible differential characteristic with 16 active S-boxes attains the DDT-product probability 2^{-32}; combining this witness with the certified active-S-box bound proves its optimality *within the specified four-round characteristic model*. Finally, the characteristic is realized by an explicit master key and plaintext pair in a four-round reduced Midori-64 implementation with key addition, round constants, and whitening. We distinguish a characteristic probability product from an exact fixed-key differential probability and document exploratory conditional fixed-key sampling near 2^{-22} for the exhibited master key. This reproduces known fixed-key phenomena rather than claiming a new security bound for the full cipher.

1 Introduction and Scope

Midori is a family of lightweight block ciphers designed for low energy consumption [1]. This study focuses on the 64-bit variant, Midori-64, whose state consists of sixteen four-bit cells. Our goal is to reproduce the four-round active-S-box bound using an exact MILP model, derive an optimal DDT-product characteristic within that model, and independently validate its four-round realization under a concrete master key.

The modeling methodology follows the S-box, permutation, and XOR construction in Ilter and Selçuk's analysis of the *FUTURE* block cipher [2]; we apply the *method*, not FUTURE's specific S-box or linear layer, to Midori-64. Foundational MILP studies include Mouha et al. [4] and the convex-hull formulation of Sun et al. [5]. All component constraints are retained unchanged between the active-S-box and probability optimizations; only the objective function is changed.

We model four consecutive SubCell layers, indexed $r = 0, 1, 2, 3$, with three intervening ShuffleCell/MixColumn layers. The last round has no following ShuffleCell/MixColumn operation. This convention also defines the four-round reduced cipher used in the verification experiment. All cell indices below follow a column-major order: cells $0, 1, 2, 3$ form the first column, and so on.

2 Preliminaries

2.1 The Midori-64 round components

The four-bit Midori-64 substitution used in this implementation is

$$S = (\mathtt{C}, \mathtt{A}, \mathtt{D}, \mathtt{3}, \mathtt{E}, \mathtt{B}, \mathtt{F}, \mathtt{7}, \mathtt{8}, \mathtt{9}, \mathtt{1}, \mathtt{5}, \mathtt{0}, \mathtt{2}, \mathtt{4}, \mathtt{6}),$$

where entries are listed in input order $0, 1, \ldots, 15$ (hexadecimal notation). Each state consists of 16 nibbles and thus 64 bits. SubCell applies S to every nibble. ShuffleCell permutes cell positions, and MixColumn combines the four cells of each column by XOR.

For a nonzero input difference Δx, an S-box is called *active*. The difference distribution table (DDT) is

$$D[\alpha, \beta] = \left|\{u \in \{0, \ldots, 15\} : S(u) \oplus S(u \oplus \alpha) = \beta\}\right|, \qquad p(\alpha, \beta) = \frac{D[\alpha, \beta]}{16}. \tag{1}$$

The computed DDT contains 159 zero entries, 72 entries equal to 2, 24 entries equal to 4, and one entry equal to 16. Thus 97 of the 256 input/output difference pairs are allowed. For an active S-box, the largest single-transition probability is $4/16 = 2^{-2}$.

2.2 Characteristic probability convention

A differential *characteristic* specifies the input and output difference of every S-box along the chosen four-round path. Its standard DDT-product weight is

$$P_{\text{trail}} = \prod_{r=0}^{3} \prod_{c=0}^{15} \frac{D[\Delta x_{r,c}, \Delta y_{r,c}]}{16}. \tag{2}$$

This is a product of local differential transition probabilities; it is not, without additional assumptions, a measured probability of the entire specified path for an individual fixed key. It is also distinct from a *differential* probability obtained by summing over multiple characteristics with matching external differences.

3 Exact MILP Formulation

3.1 S-box activity indicators

Let x_0, x_1, x_2, x_3 be the four input difference bits of a bijective S-box; let y_0, y_1, y_2, y_3 denote its output difference bits; and introduce an activity indicator $A \in \{0, 1\}$. The seven activity and nonzero-propagation inequalities used in [2, Sec. 4.1] are

$$x_i - A \leq 0 \qquad (i = 0, 1, 2, 3), \tag{3}$$

$$\sum_{i=0}^{3} x_i - A \geq 0, \tag{4}$$

$$4\sum_{i=0}^{3} x_i - \sum_{i=0}^{3} y_i \geq 0, \tag{5}$$

$$4\sum_{i=0}^{3} y_i - \sum_{i=0}^{3} x_i \geq 0. \tag{6}$$

These ensure that $A = 1$ precisely when the input difference is nonzero and that a nonzero input difference cannot produce a zero output difference (or vice versa). There are $7 \cdot 64 = 448$ such inequalities in the four-round model. They encode activity but do not, by themselves, capture the complete DDT.

3.2 Exact S-box transitions and probability costs

To encode exact S-box transitions and probabilities, we follow the H-representation and set-cover approach of Sasaki and Todo [3], as used in [2, Sec. 4.1]. Every S-box transition is represented by a ten-bit vector

$$E = (x_3, x_2, x_1, x_0, y_3, y_2, y_1, y_0, P_0, P_1).$$

The two probability bits encode $-\log_2 p$ as follows:

Table 1: Exact probability encoding for allowed Midori-64 S-box transitions.

DDT count	Local probability	P_0	P_1	$P_0 + 2P_1$
16	1	0	0	0
4	2^{-2}	0	1	2
2	2^{-3}	1	1	3

The underlying SageMath construction forms the convex hull of all 97 allowed binary vectors, computes its H-representation, and uses a set-cover MILP to select inequalities excluding every disallowed binary vector. The resulting system consists of **19 inequalities per S-box**. The precise 19 rows are provided in `selected_inequalities.txt` and are loaded unchanged by the model generator. An exhaustive verification over all $2^{10} = 1024$ binary vectors confirms that *exactly* the 97 DDT-consistent vectors are accepted; all 927 other assignments are rejected. Over 64 S-box instances, this contributes $19 \cdot 64 = 1,216$ inequalities.

The seven activity inequalities and the 19 exact-DDT inequalities are included together in *both* full-round optimization models. This shared S-box representation is central to methodological consistency.

3.3 ShuffleCell permutation constraints

For the state serialization used here, the ShuffleCell output at position c reads the input at position $\pi(c)$, where

$$\pi = (0, 10, 5, 15, 14, 4, 11, 1, 9, 3, 12, 6, 7, 13, 2, 8). \tag{7}$$

Let $y_{r,c,b}$ be bit b of the SubCell output difference at round r and $z_{r,c,b}$ the corresponding ShuffleCell output bit. The permutation is modeled exactly by the equality

$$z_{r,c,b} = y_{r,\pi(c),b}, \qquad r = 0, 1, 2; \quad c = 0, \ldots, 15; \quad b = 0, \ldots, 3. \tag{8}$$

Three layers of 64 bit equalities produce 192 constraints, in the manner of [2, Sec. 4.2].

3.4 MixColumn and XOR parity constraints

Each column is represented by four consecutive cells. With column input (z_0, z_1, z_2, z_3), the Midori MixColumn transformation is

$$\begin{pmatrix} w_0 \\ w_1 \\ w_2 \\ w_3 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{pmatrix} \begin{pmatrix} z_0 \\ z_1 \\ z_2 \\ z_3 \end{pmatrix} \quad \text{over } \mathbb{F}_2^4. \tag{9}$$

In particular, $w_0 = z_1 \oplus z_2 \oplus z_3$. For three input bits a, b, c and output bit $d = a \oplus b \oplus c$, the parity is encoded by

$$a + b + c + d = 4u - 2v, \qquad u, v \in \{0, 1\}. \tag{10}$$

The output of each inter-round MixColumn becomes the next SubCell input $x_{r+1,c,b}$. Using (10) for every bit gives 192 parity equalities and 384 auxiliary binary variables. Exhaustive testing of all 16 assignments (a, b, c, d) confirms that the equation is feasible in the dummy variables if and only if $d = a \oplus b \oplus c$.

4 Four-Round Model and Experimental Setup

4.1 A common feasible region for both objectives

Let $A_{r,c}$ denote S-box activity and $P_{0,r,c}, P_{1,r,c}$ its probability bits. We use the same four-round constraints for both experiments and require a nontrivial initial difference:

$$\sum_{c=0}^{15}\sum_{b=0}^{3} x_{0,c,b} \geq 1. \tag{11}$$

Only the objective changes:

$$\text{Active S-box count: } \min N_{\text{active}} = \min \sum_{r=0}^{3}\sum_{c=0}^{15} A_{r,c}, \tag{12}$$

$$\text{Characteristic cost: } \min C = \min \sum_{r=0}^{3}\sum_{c=0}^{15} (P_{0,r,c} + 2P_{1,r,c}) = \min(-\log_2 P_{\text{trail}}). \tag{13}$$

Table 2: Size of each four-round MILP before presolve.

Constraint category	Number
S-box activity inequalities (64×7)	448
S-box H-representation inequalities (64×19)	1,216
ShuffleCell bit equalities (3×64)	192
MixColumn XOR equalities (3×64)	192
Nonzero initial difference	1
Total constraints	**2,049**
Total binary variables	**1,280**

The 1,280 variables consist of 512 SubCell input/output difference bits, 64 activity indicators, 128 probability bits, 192 ShuffleCell bits, and 384 XOR dummy bits. The LP generator writes separate files for the two objectives while keeping their *constraint systems identical*.

4.2 Validation and solver environment

The component tests check the entire ten-bit H-representation truth table, the XOR parity encoding, the agreement of the two models' constraint systems, and randomly constructed valid differential trails. Fourteen automated tests pass, including a replay of the published experiment records. Independently, the saved Gurobi activity solution was checked against all 2,049 inequalities with zero violations; the probability trail was checked against a separately implemented Midori DDT and linear layers.

Optimization used Gurobi 13.0.3 on Windows 11 with an AMD Ryzen AI 7 350 CPU. The final activity-count run used eight threads, `MIPFocus=3`, `MIPGap=0`, and a six-hour time limit. A previously obtained feasible probability-trail solution with objective 16 was supplied as a MIP start; the new run *certified optimality*, rather than discovering this incumbent. The solver terminated well before the time limit.

5 Certified Minimum Number of Active S-boxes

Gurobi *certified* a minimum of 16 active S-boxes for the four-round MILP described in Section 4. The run loaded a feasible 16-active solution as a MIP start (recorded as `Loaded user MIP start`

`with objective 16`). The solver's contribution was to close the lower bound to 16 and certify global optimality within the model; it did not discover the initial incumbent during this run. The final solver information is summarized in Table 3.

Table 3: Certified four-round active-S-box result (Gurobi 13.0.3).

Solver status	`OPTIMAL`
Best feasible objective (incumbent)	16
Best objective bound	16
Relative optimality gap	0%
Branch-and-bound nodes explored	60,345
Runtime	794.455 s ($\approx$ 13 min 14 s)
Model size	2,049 constraints; 1,280 binary variables

Because the solver's certified lower bound matches the feasible incumbent,

$$\boxed{\min N_{\text{active}} = 16} \tag{14}$$

within this exact four-round model (subject to standard numerical tolerances). This independently reproduces the four-round value reported for Midori64 by its designers [1]. It is *not* a newly discovered active-S-box lower bound for the cipher.

An optimal solution has the following active cell indices, numbered from 0 through 15:

Table 4: Activity pattern of the saved optimal solution.

Round	Active cell indices	Count
1	$0, 10, 15$	3
2	2	1
3	$12, 13, 15$	3
4	$0, 1, 2, 8, 9, 11, 12, 14, 15$	9
Total		**16**

5.1 The wide-trail bound and independent word-level validation

The theoretical explanation predates this implementation. Midori64's MixColumn has differential branch number $B = 4$, and its ShuffleCell is an optimal cell permutation for diffusion. Under these conditions, the four-round propagation theorem of Daemen and Rijmen [6, Ch. 9, Sec. 9.5] gives at least $B^2 = 16$ active S-boxes across four rounds. The original Midori paper reports this value in its active-S-box comparison (Table 8) [1]. The following dynamic program provides a separate, directly executable verification of the bound for Midori's *actual* column transformation and cell permutation, without invoking a floating-point MILP solver.

A separate verification does not require MILP or numerical solver tolerances. Represent an active-cell pattern by a nonzero binary word $u \in \{0,1\}^{16}$ and let $w(u)$ be its Hamming weight. For each four-cell input pattern to MixColumn, enumerate the possible nonzero output patterns by applying the exact linear transformation to all 16^4 nibble vectors. Let T denote the induced, four-column relation, composed with the ShuffleCell permutation. Then the dynamic program

$$F_1(u) = \begin{cases} w(u), & u \neq 0, \\ +\infty, & u = 0, \end{cases} \qquad F_{r+1}(v) = w(v) + \min_{u:\, u\,T\,v} F_r(u)$$

minimizes the sum of active-cell counts over the *relaxed* activity model. Because every concrete bit-level characteristic induces a permissible sequence of activity patterns, its active-S-box count is

at least $\min_v F_4(v)$. The independently implemented program `independent_checks.py` produces, for one through eight rounds,

$$(1, 4, 7, 16, 23, 30, 35, 38).$$

In particular, $\min_v F_4(v) = 16$, and our explicit bit-level 16-active characteristic attains this lower bound. This complements the Gurobi calculation with a combinatorial, solver-independent lower bound. Together with the explicit 16-active bit-level characteristic, it establishes equality in the stated four-round setting. This reproduces a known diffusion result; it is not a new cryptanalytic bound.

5.2 An equivalent compact LP

The 192 ShuffleCell bit equalities can be eliminated by substituting $z_{r,c,b} = y_{r,\Pi(c),b}$ directly into the MixColumn equations. The resulting LP has 1,857 constraints and 1,088 binary variables. An automated equivalence audit checks that the archived full-model optimum projects onto a feasible compact-model assignment of objective 16. The compact construction is designed to fit Gurobi's standard 2,000-row, 2,000-variable size-limited license; the archived `OPTIMAL` solver log is nevertheless for the original full model.

6 Best Four-Round Differential Characteristic

A separate probability-minimizing Gurobi run found the characteristic in Table 5. The direct probability optimization was interrupted after $2,289.24$ seconds: its best cost was 32, its branch-and-bound lower bound was 18, and its reported gap was 43.75%. Therefore, *that run alone did not certify probability optimality.* The certified active-S-box result supplies the missing bound on the DDT-product weight.

Table 5: Four-round Midori-64 differential characteristic; states are 16 hex nibbles in column-major order.

Round	SubCell input Δx	SubCell output Δy	Active	p_r
1	2000000000200002	4000000000400004	3	2^{-6}
2	0040000000000000	0020000000000000	1	2^{-2}
3	0000000000002202	0000000000004404	3	2^{-6}
4	4440000044044044	2220000022022022	9	2^{-18}
Total			**16**	$\mathbf{2^{-32}}$

Each of the 16 active transitions in this path has a DDT count of 4 and therefore probability 2^{-2}. Inactive S-boxes contribute probability 1. Consequently,

$$P_{\text{trail}} = 2^{-6} \cdot 2^{-2} \cdot 2^{-6} \cdot 2^{-18} = \boxed{2^{-32}}, \qquad C = 32. \tag{15}$$

The reported SubCell outputs propagate exactly to the following round's inputs under ShuffleCell and MixColumn. This was independently checked using a reference implementation rather than trusting the MILP encoding alone.

Proposition 1 (Characteristic optimality in the specified MILP model)**.** *Within the four-round model of Sections 2–4, the maximal DDT-product probability of a nontrivial differential characteristic is* 2^{-32}.

Proof. The active-S-box minimization certifies that every feasible nontrivial four-round characteristic has at least 16 active S-boxes. The maximal DDT probability for each active Midori-64 S-box transition is 2^{-2}, while an inactive transition has probability 1. Thus, for every such characteristic,

$$P_{\text{trail}} \leq (2^{-2})^{N_{\text{active}}} \leq 2^{-32}.$$

The characteristic in Table 5 achieves 2^{-32} by (15), so the bound is attained. □

This proposition establishes optimality for *single characteristics in the modeled four-round construction*. It does not assert an optimum for the full 16-round cipher or for differential probabilities aggregated over trails.

7 Independent Software Verification

7.1 Layer-by-layer validation

A separately implemented four-round Midori core recomputed the DDT and checked all 64 local nibble transitions. The specified differences are valid and the four active-S-box counts are $(3, 1, 3, 9)$. Independent ShuffleCell/MixColumn operations map each of the first three SubCell output differences exactly to the next SubCell input difference. These checks pass. However, locally valid S-box transitions do not on their own demonstrate that one plaintext pair realizes the entire multi-round path.

7.2 Constructive plaintext-pair verification

A second MILP held the four-round difference trail fixed and searched for a 128-bit master key and a pair of plaintexts realizing every internal transition. The reduced Midori-64 implementation uses $K = K_0 \| K_1$, the whitening key $WK = K_0 \oplus K_1$, three round-key additions with

$$RK_i = K_{i \bmod 2} \oplus \alpha_i \qquad (i = 0, 1, 2), \tag{16}$$

and initial and final whitening. The fixed α_i are the first three Midori-64 round constants in the model's state serialization. The configuration has three full rounds and a final SubCell, hence *four SubCell rounds in total*; it is not a full 16-round encryption.

The search returned the master key

$$K = \texttt{00000101000100016177166671066660},$$

whitening key $WK = \texttt{6177176771076661}$, and round keys

$$RK_0 = \texttt{0001000010100010},$$
$$RK_1 = \texttt{6066066660066660},$$
$$RK_2 = \texttt{1010000100100100}.$$

With this master key, the following explicit plaintexts and ciphertexts were obtained:

Table 6: Constructive four-round witness under the displayed master key.

	First text	Second text
Plaintext	`ADBBDBABBDCBAAAD`	`8DBBDBABBDEBAAAF`
Ciphertext	`ADBDDBADBDABACAD`	`8F9DDBAD9FA98C8F`

In particular,

$$\Delta P = \texttt{2000000000200002},$$
$$\Delta C = \texttt{2220000022022022}.$$

Independent execution of both plaintexts through the reduced-round implementation reproduces *every* SubCell input/output difference in Table 5, not merely the external differences. The result is

an explicit end-to-end witness of feasibility for one standard-schedule master key in the specified reduced four-round variant. For comparison, this exact trail was infeasible in a separate witness search where the master key was fixed to all zeros; realizability of an exact path can depend on the key.

The witness MILP had an identically zero objective. Its solver status `OPTIMAL` means that a feasible pair was found; this must not be confused with optimization of the differential characteristic. Finally, one witness does not measure a 2^{-32} empirical occurrence rate for the fixed-key encryption permutation; a separate conditional sampling experiment for this key is reported in Section 8.

8 Exploratory Fixed-Key Characteristic Behaviour

The 2^{-32} weight computed from local DDT entries is not the fixed-key probability of the specified path. In particular, both active transitions used in the witness involve affine conforming-input sets:

$$2 \to 4 : \ \{\mathtt{C}, \mathtt{D}, \mathtt{E}, \mathtt{F}\}, \qquad 4 \to 2 : \ \{0, 2, 4, 6\}.$$

The $2 \to 4$ conforming outputs lie in the XOR subgroup $E = \{0, 2, 4, 6\}$. At the sole active S-box of round two,

$$x_{1,2} = S(x_{0,0}) \oplus S(x_{0,10}) \oplus S(x_{0,15}) \oplus RK_{0,2}.$$

Consequently, the round-two transition is compatible with all conforming first-round triples if $RK_{0,2} \in E$, and incompatible with all such triples otherwise. A uniformly chosen master key satisfies this necessary condition with probability $1/4$. This is not a complete characterization of the keys for which all four rounds can follow the path.

A separate C implementation performs conditional sampling for the specific verified master key: it chooses first-round active inputs uniformly among the conforming triples and samples the other nibbles. The exact conditioning probability 2^{-6} is incorporated when estimating the full-path probability. Two runs with 10^8 conditional samples each yielded 1,541 and 1,543 hits, corresponding to estimated probabilities approximately $2^{-21.99}$ and $2^{-21.98}$. Thus the observed fixed-key path probability is near 2^{-22}, well above the DDT-product 2^{-32}. These are statistical estimates, not an analytic probability or a claim about the fraction of compatible master keys. We interpret them in the established fixed-key and plateau-characteristic literature, in particular Canteaut and Fruchon [7]. The underlying phenomenon is not claimed as new.

9 Discussion, Limitations, and Conclusion

This reproducible implementation recovers the known four-round Midori-64 minimum of 16 active S-boxes by two independent methods: Gurobi's `OPTIMAL` certificate for the exact bit-level MILP, and a solver-free word-level dynamic program. The latter is consistent with the classical wide-trail four-round propagation bound. Because the maximum probability of a nonzero Midori S-box transition is 2^{-2}, every nontrivial four-round characteristic has a DDT-product of at most 2^{-32}. The explicit characteristic in Table 5 attains this bound. Its realizability is confirmed by a master key and plaintext pair under the standard key schedule in a reduced four-round Midori-64 implementation.

The DDT-product is not the actual occurrence probability of the same path under every fixed master key. Conditional sampling for the displayed master key estimates a path probability near 2^{-22}, whereas the round-two algebraic condition in Section 8 already excludes three quarters of uniformly chosen keys. This illustrates fixed-key/plateau behaviour studied in the existing literature, rather than a new attack or security result. Deriving the complete set of compatible-key conditions and an exact fixed-key probability is a natural direction for further investigation. The present contribution is the transparent modeling, exhaustive component tests, independent lower-bound calculation, saved optimization record, and constructive software verification of a known four-round benchmark.

A Reproducibility and Saved Artifacts

The model builder (`build_models.py`) and solver (`solve.py`) share the saved inequalities. Software verification uses `midori_reference.py` and `verify_trail.py`. Additional scripts perform focused active-S-box optimization, an independent constraint audit, and a standard-key witness search. In the repository root, the principal commands are:

```
python build_models.py
python -u -m unittest discover -s tests -v
python -u overnight_midori_active.py --hours 6
python -u solve.py --objective probability --time-limit 7200
python independent_checks.py
python compact_models.py
python -u verify_trail.py --report results/probability_trail.json --samples 0
python -u find_midori_standard_key_pair.py --report results/probability_trail.json --search-
    master-key --time-limit 120
```

The retained experiment records and reproducibility tools provide the audit trail:

- `results/active_optimality.json` — certified 16/16 objective and bound, status `OPTIMAL`;
- `results/active_solver.log` — full branch-and-bound record;
- `results/active_solution.sol` — the complete 1,280-variable assignment;
- `results/probability_trail.json` — objective 32 and all four rounds' difference states;
- `results/standard_key_witness.json` — concrete master key, plaintexts, ciphertexts, and witness check;
- `results/fixed_key_sampling.json` — conditional Monte Carlo settings, seeds, and observed hits.

The probability-only run was interrupted and does not have solver-certified optimality. The characteristic optimum follows instead from the certified active-S-box bound and Proposition 1. Numerical solver certification is subject to Gurobi tolerances, as in standard MILP practice. A word-level dynamic program independently provides the same 16-active lower bound. The archived standard-key witness is separately replayed against a test-vector-validated Midori implementation. The repository also provides a small feasibility MILP to obtain a fresh witness, which may differ from the recorded one.

References

[1] S. Banik, A. Bogdanov, T. Isobe, K. Shibutani, H. Hiwatari, T. Akishita, and F. Regazzoni. Midori: A block cipher for low energy. In *Advances in Cryptology—ASIACRYPT 2015*, Part II, LNCS 9453, pp. 411–436, Springer, 2015. https://eprint.iacr.org/2015/1142.

[2] M. B. İlter and A. A. Selçuk. MILP-aided cryptanalysis of the FUTURE block cipher. In *Innovative Security Solutions for Information Technology and Communications (SecITC 2022)*, LNCS 13809, pp. 153–167, Springer, 2023. https://doi.org/10.1007/978-3-031-32636-3_9. Preprint: https://eprint.iacr.org/2022/1398.

[3] Y. Sasaki and Y. Todo. New algorithm for modeling S-box in MILP based differential and division trail search. In *Innovative Security Solutions for Information Technology and Communications (SecITC 2017)*, LNCS 10543, pp. 150–165, Springer, 2017. https://doi.org/10.1007/978-3-319-69284-5_11.

[4] N. Mouha, Q. Wang, D. Gu, and B. Preneel. Differential and linear cryptanalysis using mixed-integer linear programming. In *Inscrypt 2011*, LNCS 7537, pp. 57–76, Springer, 2012. https://doi.org/10.1007/978-3-642-34704-7_5.

[5] S. Sun, L. Hu, P. Wang, K. Qiao, X. Ma, and L. Song. Automatic security evaluation and (related-key) differential characteristic search. In *ASIACRYPT 2014*, LNCS 8873, pp. 158–178, Springer, 2014. https://eprint.iacr.org/2013/676.

[6] J. Daemen and V. Rijmen. *The Design of Rijndael: AES—The Advanced Encryption Standard*. Springer, 2002, Chapter 9, Section 9.5 (Four-Round Propagation Theorem). https://doi.org/10.1007/978-3-662-04722-4.

[7] A. Canteaut and M. Fruchon. Understanding unexpected fixed-key differential behaviours: How to avoid major weaknesses in lightweight designs. *ASIACRYPT 2025*. Extended version: https://eprint.iacr.org/2025/1553.